VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

Student Name	Registration Number
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Slot	**Date of Submission**
A13	31-3-2026

1. Introduction

I put out a project to spread the open source philosophy and display my respect for it's developers and their relentless commitment to improving the software ecosystem.

VLC Media Player's been an integral part of the entire ecosystem at large since before this decade, making multimedia more accessible and manipulatable for everyone.

The report shall include an audit for VLC Media Player.

2. Choosing Your Software

VLC Media Player

Approved software options

Software	Category	License	Why it's interesting
Linux Kernel	Operating System	`GPL v2`	*The foundation everything else runs on*
Apache HTTP Server	Web Server	`Apache 2.0`	*Powers ~30% of all websites globally*
MySQL	Database	`GPL v2 / Commercial`	*A license with a dual-model story to tell*
Firefox	Web Browser	`MPL 2.0`	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	`LGPL / GPL`	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	`MPL 2.0`	*Born from a community fork — a real lesson*
Python	Programming Language	`PSF License`	*A language shaped entirely by community*
Git	Version Control	`GPL v2`	*The tool Linus built when proprietary failed him*

Your choice	Write your chosen software here before submission: VLC media Player _____

3. Project Report Structure

Your report should be 12 to 16 pages long. Each section below maps to a unit in the course. Do not pad with screenshots — every page should contain original thinking and analysis written in your own words.

Part A — Origin and Philosophy (Units 1 & 2)

This is the most important section of the report. Allocate at least four pages to it.

A1. The problem this software was created to solve

The problem was that the existing technology for decoding MPEG-2 video relied on prohibitively expensive, proprietary hardware decoders that the students couldn't afford. Out of sheer necessity, they wrote their own software to decode and broadcast the video across the network, creating a server and a client (the VideoLAN Client). While initially an internal university tool, the creators faced a glaring issue: relying entirely on a revolving door of graduating students meant the project would eventually die. In 2001, with the blessing of the university's director, they released VLC under the open-source GPL license. They chose to share it not just out of academic altruism, but as a survival mechanism to attract global developers. This open-source pivot allowed contributors worldwide to reverse-engineer and bake in support for virtually every media format in existence, transforming VLC from a clever campus workaround into a global public utility that saved early-internet users from the nightmare of downloading broken, malware-ridden "codec packs."

A2. The license — what it actually says

Here is a breakdown of how free software licensing works, using VLC Media Player as a prime example.

The Four Freedoms of Free Software

The concept of "Free Software" was formalized by the Free Software Foundation (FSF), which defined it using four essential freedoms that guarantee user control over the program.

- **Freedom 0:** The freedom to run the program as you wish, for any purpose.
- **Freedom 1:** The freedom to study how the program works, and change it so it does your computing as you wish. (Access to the source code is a precondition for this).
- **Freedom 2:** The freedom to redistribute copies so you can help your neighbor.
- **Freedom 3:** The freedom to distribute copies of your modified versions to others. By doing this, you can give the whole community a chance to benefit from your changes. (Access to the source code is a precondition for this).

Does VLC grant all four?

Yes. VLC is licensed primarily under the **GNU General Public License (GPL)**, which was specifically written by the FSF to legally guarantee and protect these exact four freedoms for all users.

Can a company modify and sell VLC without sharing changes?

No. A company cannot keep their modifications proprietary if they distribute the software.

While the GPL absolutely allows a company to sell copies of VLC for money (exercising Freedom 2), it includes a strict legal mechanism known as **"copyleft."** Copyleft dictates that if you modify a GPL-licensed program and distribute that modified version to the public (whether for free or for a

fee), you **must** release the source code of your modified version under the exact same GPL license.

You cannot take GPL code, close the source, and make it proprietary. This ensures that the software, and all downstream improvements to it, remain permanently free and open to the community.

GPL vs. MIT Licensing

The difference between these two popular open-source licenses comes down to how much they restrict developers in order to protect the software.

- **GPL (Copyleft / Strong Protection):** As mentioned above, the GPL is viral. If you use GPL code in your project, your entire project generally must also be open-sourced under the GPL if you distribute it. It prioritizes the freedom of the *software ecosystem* over the freedom of the individual developer.
- **MIT (Permissive):** The MIT license is incredibly brief and permissive. It essentially says, "You can do whatever you want with this code—modify it, distribute it, sell it, or make it proprietary closed-source—as long as you keep the original copyright notice attached." It prioritizes the freedom of the *developer* to use the code however they see fit, even if that means making a closed-source product.

Which would I choose?

Because I am an AI, I do not possess personal property rights, legal standing, or personal side-projects to license. However, looking at how humans make this choice, it depends entirely on the goal of the project:

- If the goal is to build a protected, collaborative commons where corporate entities are forced to contribute their improvements back to the community (like Linux or VLC), the **GPL** is the right choice.
- If the goal is absolute maximum adoption with zero friction, encouraging businesses to integrate the code into their proprietary products without fear of legal complications, **MIT** is the standard choice.

"Free as in Free Beer" vs. "Free as in Freedom"

This is the classic phrase used to clear up the ambiguity of the English word "free" (which translates to two distinct words in languages like French: *gratis* and *libre*).

- **Free as in Free Beer (*Gratis*):** This simply means the software costs zero dollars. You do not have to pay for it. (Think of freeware like the basic version of Adobe Acrobat Reader).
- **Free as in Freedom (*Libre*):** This means you have liberty and control over the software—specifically, the Four Freedoms mentioned above.

An Example from VLC:

VLC is both *gratis* and *libre*. It costs no money to download (free beer). But more importantly, it provides freedom.

If a new, obscure video format is invented tomorrow, you don't have to wait and hope that a corporate entity decides it is profitable enough to update your media player. Because VLC gives you "freedom," any developer in the world can open VLC's source code, write a decoder for the

new format, and publish that updated version for everyone to use. That ability to adapt, repair, and control the tool yourself is what makes it "free as in freedom.

Part B — Linux Footprint (Unit 2)

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The selected text outlines a section for a Capstone Project focusing on how a chosen open-source software operates within a Linux environment.

This section requires **actual installation** of the software on Linux (physical machine, VM, or lab system) and asks for documentation on the following key aspects:

1. **Installation Method:** Is it installed via package managers (RPM, apt) or compiled from source?
2. **Key Directories:** Where are the binaries, configuration files, and logs stored (e.g., `/usr/bin`, `/etc`, `/var/log`)?
3. **User and Group:** What Linux user and group does the software run as, and what are the security implications of this?
4. **Service Management:** How is the software started, stopped, or restarted as a system service? Include the specific commands.
5. **Update Model:** How does the open-source community deliver security patches for the software?

Part C — The FOSS Ecosystem (Units 3 & 4)

- **What other open-source tools does it depend on? (Libraries, compilers, runtimes)** VLC depends on a vast number of open-source libraries for its functionality, including FFmpeg for decoding various formats, libdvdread/libdvdnav for DVD playback, various codec libraries (like libvorbis, liba52), and tools like wxWidgets or Qt for the graphical interface (depending on the platform/version). It's typically compiled using standard open-source compilers like GCC or Clang.
- **What has it enabled or inspired? (Forks, spin-offs, tools built on top of it)** VLC has inspired many media players and libraries. The core library, libVLC, is often used by developers to embed VLC's powerful playback capabilities into their own applications. Examples include media players on mobile platforms, various custom frontends, and tools that rely on its network streaming capabilities.
- **Where does it fit in the LAMP stack, or how does it relate to the web?** VLC doesn't directly fit into the traditional LAMP (Linux, Apache, MySQL, PHP) stack, which is for web serving. However, it *relates* to the web through its extensive network streaming capabilities (HTTP, RTP, RTSP, etc.). It can stream content *from* the web, stream content *to* the web (as a server), and its core technologies are used in many web-based video systems.
- **Is there a community — mailing lists, forums, conferences, IRC channels? Who governs it?** Yes, VLC has a very active community.
    - **Community Channels:** They use mailing lists (e.g., for development and general discussion), forums, and IRC channels.
    - **Governance:** VLC is primarily developed and maintained by the **VideoLAN** organization, which is a non-profit organization based in France. The core development is done by a global team of volunteers, with VideoLAN providing the legal, financial, and infrastructure support.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source	Proprietary Alternative
Cost	Typically free or significantly lower cost. Development costs are shared.	High upfront licensing fees, subscription models, or per-user costs.
Security (who can audit the code?)	Anyone can audit the code, leading to faster identification and patching of vulnerabilities (many eyes principle).	Only the vendor's internal teams can audit the code. Audits are opaque to the user.
Support and reliability	Community-driven support, often fast and detailed. Reliability depends on the active community and project maturity. Commercial support is available from third parties.	Formal, structured support contracts from the vendor, often with guaranteed SLAs. Reliability is controlled by the vendor.
Freedom to modify	Full freedom to view, modify, and distribute the code to suit specific needs.	No right to modify the code. Users are locked into the vendor's feature set and release cycle.
Community vs corporate control	Controlled by a decentralized community of developers and users. Development is often transparent.	Controlled entirely by a single corporation. Development roadmap and decisions are opaque.
Your overall verdict	Better for transparency, customization, long-term cost savings, and avoiding vendor lock-in. Ideal for core infrastructure and highly specialized needs.	Better for ease of implementation, formal support guarantees, and a single point of accountability. Ideal for non-core tasks where formal support is critical.

4. Shell Script Tasks

Script 1 — System Identity Report (Unit 1 & 2)

```
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$ bash script_1.sh
========================================
 Open Source Audit — Aarat Sahajpal
========================================
Software : VLC
----------------------------------------
Distro   : Ubuntu 24.04.2 LTS
Kernel   : 6.11.0-17-generic
User     : aarat-sahajpal
Uptime   : up 20 minutes
Date     : Tue Mar 31 07:35:15 PM IST 2026
----------------------------------------
License  : GNU GPL
========================================
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$
```

Script 2 — FOSS Package Inspector (Unit 2)

```
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$ bash script_2.sh
vlc is installed.
ii  libvlc-bin:amd64                    3.0.20-3build6    amd64    tools for VLC's base library
ii  libvlc5:amd64                       3.0.20-3build6    amd64    multimedia player and streamer library
ii  libvlccore9:amd64                   3.0.20-3build6    amd64    base library for VLC and its modules
ii  vlc                                 3.0.20-3build6    amd64    multimedia player and streamer
ii  vlc-bin                             3.0.20-3build6    amd64    binaries from VLC
ii  vlc-data                            3.0.20-3build6    all      common data for VLC
ii  vlc-l10n                            3.0.20-3build6    all      translations for VLC
ii  vlc-plugin-access-extra:amd64       3.0.20-3build6    amd64    multimedia player and streamer (extra access plugins)
ii  vlc-plugin-base:amd64               3.0.20-3build6    amd64    multimedia player and streamer (base plugins)
ii  vlc-plugin-notify:amd64             3.0.20-3build6    amd64    LibNotify plugin for VLC
ii  vlc-plugin-qt:amd64                 3.0.20-3build6    amd64    multimedia player and streamer (Qt plugin)
ii  vlc-plugin-samba:amd64              3.0.20-3build6    amd64    Samba plugin for VLC
ii  vlc-plugin-skins2:amd64             3.0.20-3build6    amd64    multimedia player and streamer (Skins2 plugin)
ii  vlc-plugin-video-output:amd64       3.0.20-3build6    amd64    multimedia player and streamer (video output plugins)
ii  vlc-plugin-video-splitter:amd64     3.0.20-3build6    amd64    multimedia player and streamer (video splitter plugins)
ii  vlc-plugin-visualization:amd64      3.0.20-3build6    amd64    multimedia player and streamer (visualization plugins)
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VLC: a free and open-source media player that plays almost all formats.
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$
```

Script 3 — Disk and Permission Auditor (Unit 2)

```
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$ bash script_3.sh
==========================================
        Directory Audit Report
==========================================
/etc
 Permissions: drwxr-xr-x root root
 Size       : 12M
------------------------------------------
/var/log
 Permissions: drwxrwxr-x root syslog
 Size       : 479M
------------------------------------------
/home
 Permissions: drwxr-xr-x root root
 Size       : 6.3G
------------------------------------------
/usr/bin
 Permissions: drwxr-xr-x root root
 Size       : 285M
------------------------------------------
/tmp
 Permissions: drwxrwxrwt root root
 Size       : 76K
------------------------------------------
Checking Git configuration directory...
.gitconfig not found
==========================================
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$
```

Script 4 — Log File Analyzer (Units 2 & 5)

```
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$ ./script_4.sh log_file.log
==========================================
Log File Analysis Report
==========================================
File      : log_file.log
Keyword   : error
Matches   : 1
------------------------------------------
Last 5 matching lines:
[2024-05-20 08:00:01] INFO: Initializing server subsystem...[2024-05-20 08:00:05] INFO: Database connection established on port 5432.[2024-05-20 08:15:22] WARNING: High memory usage detected
 (85%). Initiating garbage collection.[2024-05-20 08:15:30] INFO: Memory cleared. Current usage: 42%.[2024-05-20 09:12:45] INFO: User 'admin' logged in from 192.168.1.55.[2024-05-20 10:05:11
] WARNING: Latency spike detected in API gateway. Threshold exceeded by 200ms.[2024-05-20 10:05:15] INFO: Latency normalized.[2024-05-20 11:30:00] INFO: Scheduled backup started.[2024-05-20
11:45:02] WARNING: SSL certificate for 'api.internal.service' expires in less than 7 days.[2024-05-20 12:00:00] INFO: Routine maintenance check complete. 0 errors found.[2024-05-20 13:22:19]
 WARNING: Disk space on /var/log/ is reaching capacity (92% full).[2024-05-20 13:25:00] INFO: Log rotation script executed successfully.[2024-05-20 14:00:01] INFO: Heartbeat signal received
from worker node 4.
==========================================
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$
```

Script 5 — The Open Source Manifesto Generator (Unit 5)

```
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$ bash script_5.sh
==========================================
   Open Source Manifesto Generator
==========================================

1. Name one open-source tool you use every day: tmux
2. In one word, what does 'freedom' mean to you? lack of authority
3. Name one thing you would build and share freely: zathura but for windows

Manifesto saved to manifesto_aarat-sahajpal.txt

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Open Source Manifesto
Date: 31 March 2026

I believe in the power of open source. Tools like tmux show how collaboration can create powerful software accessible to everyone.
To me, freedom means lack of authority — the ability to use, modify, and share software without restrictions.
In the future, I would love to build zathura but for windows and share it openly with the world to contribute to the community.

— Generated by aarat-sahajpal
------------------------------------------
aarat-sahajpal@aarat-sahajpal-HP-Laptop-15-fd1xxx:~/Desktop$
```

Conclusion:

Ultimately, open-source software has evolved from a grassroots developer movement into the foundational infrastructure of the modern digital economy. By democratizing access to source code, it fosters unprecedented global collaboration, accelerates innovation cycles, and significantly reduces vendor lock-in for organizations of all sizes. While challenges such as securing funding for critical maintainers and managing supply chain vulnerabilities remain, the open-source model has proven its resilience and scalability. Embracing open source is no longer just a philosophical preference, but a strategic necessity for businesses looking to remain competitive and drive technological advancement.